Companion document: partial results, conjectures, and computational evidence

Supplement to *The Universal Right-Triangle Reflection Sequence*

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Preface. This companion document collects partial results, conjectures, and computational evidence supporting the program of the main paper `paper.tex`. **Nothing here is claimed as a theorem.** All results are explicitly conditional or empirical: each is labelled either as a *conjecture* (a target statement we believe but have not proved), as a *conditional proposition* (a deduction valid under a hypothesis that is itself only empirically verified at a finite horizon), or as a *numerical observation* (a finite computational record). Cross-references to theorems of the main paper use the same labels.

1. The transcendence program for β_2: conjecture and three attack routes

Define the asymptotic ratio

$$\beta_2 \; := \; \lim_{d\to\infty} \frac{u_{d+1}}{u_d}$$

of the universal right-triangle reflection sequence. The limit exists and equals

$$\beta_2 \; = \; 1.4916177871143742268\ldots \; = \; 1/\sqrt{q^*},$$

where $q^* = 0.44945363055894\ldots$ is the least positive root of $\Sigma_1(q) = 1$; this is unconditional, and the singularity is a simple pole ($\Sigma_0(q^*) \neq 0$). Only the arithmetic nature of β_2 remains open.

Conjecture 1 (Transcendence of β_2)**.** *β_2 is transcendental over $\mathbb{Q}$, and the generating function $U(t) = \sum_{d\geq 0} u_d t^d$ is transcendental over $\mathbb{Q}(t)$.*

We collect three independent attack routes. Each rests on a different finite-horizon hypothesis that is empirically verified and structurally attackable; the routes are logically independent and would each individually settle Conjecture 1.

1.1 Route 1: Christol mod-p via the explicit cyclic-module structure

Let $\rho_{\text{abs}} : W \to Q \ltimes M$ be the universal realisation of the main paper (§4) with $Q = D_\infty \times \mathbb{Z}/2$ and $M = \mathbb{Z}[Q]/(Y+1, c+1)$. For each prime p, write

$$S_p \; := \; \{\, (\pi(g),\, m \bmod p) \; : \; (g,m) = \rho_{\text{abs}}(w),\; w \in W \,\} \; \subset \; Q \times (M/pM).$$

Proposition 1 (Mod-p state space is infinite)**.** *For every prime p, $|S_p| = \infty$. In particular the infinite family $\{((XY)^n, t_n) : n \geq 1\}$ (with t_n the translation part of $\rho_{\text{abs}}((R_0R_2)^n) \bmod p$) consists of pairwise distinct elements of $Q \times (M/pM)$.*

Sketch. XY has infinite order in D_∞, so the Q-components $(XY)^n$ are pairwise distinct. This alone suffices for infinity of S_p via projection. The stronger separation on the (M/pM)-coordinate, used in Route 1 below, follows from the witness monomial $(XY)^{n-1}X \cdot h$ surviving reduction modulo p by the cyclic-module relations $(Y+1)h = (c+1)h = 0$.

Numerical record. An exact-arithmetic Rust BFS in $\mathbb{F}_p[Q]$ enumerates $S_p(d) := \{\rho_{\text{abs}}(w) \bmod p : |w|_W \leq d\}$ at three primes:

p	$d_{\max}$	$\lvert S_p(d_{\max})\rvert$	β_p (ratio fit, last 10 depths)
3	42	65 428 964	1.4681
5	40	62 736 544	1.5048
7	40	65 724 148	1.5077

The implementation uses the fixed-degree truncation justified by $(Y+1)h = (c+1)h = 0$, which bounds the support of any reachable $m \in M/pM$ by $O(d)$ monomials.

Proposition 2 (β_2 transcendence: Christol route, conditional)**.** *Conditional on the Myhill–Nerode separation hypothesis (the states associated to $\{((XY)^n, t_n)\}_{n\geq 1}$ in the p-DFAO for $u_d \bmod p$ are pairwise distinct future-equivalence classes), $U(t)$ is transcendental over $\mathbb{Q}(t)$ and β_2 is transcendental over $\mathbb{Q}$.*

Sketch. Under the separation hypothesis the p-DFAO has infinitely many states, hence (Allouche–Shallit Thm. 6.6.2) the p-kernel of $U(t) \bmod p$ is infinite, so by Christol 1979 $U(t) \bmod p$ is not algebraic over $\mathbb{F}_p(t)$. Adamczewski–Bell 2017 lifts this to non-algebraicity of $U(t)$ over $\mathbb{Q}(t)$, and Flajolet–Sedgewick (*Analytic Combinatorics* Thm. VII.7) implies β_2 transcendental.

1.2 Route 2: D-finite exclusion

Proposition 3 (No D-finite recurrence at low complexity)**.** *On $u_0, \ldots, u_{42}$, no D-finite (P-recursive) recurrence of order $k \leq 9$ with polynomial-in-d coefficients of degree $m \leq 7$ exists (47 tested (k,m)-pairs, exact integer linear algebra modulo the Mersenne prime $2^{61}-1$).*

Proposition 4 (β_2 transcendence: D-finite route, conditional)**.** *Conditional on extending Proposition 3 from $(k \leq 9, m \leq 7)$ to all $(k,m) \in \mathbb{N}^2$, β_2 is transcendental over $\mathbb{Q}$.*

Sketch. By Stanley *Enumerative Combinatorics* Vol. 2 Thm. 6.4.6, every power series algebraic over $\mathbb{Q}(t)$ is D-finite. Excluding D-finite at all (k,m) excludes algebraic, hence (Flajolet–Sedgewick VII.7) transcendence of β_2. The natural structural target for the extension is a Gröbner-basis argument on the skew-polynomial ring $\mathbb{Q}\langle n, S\rangle$ acting against the cyclic-module relations $(Y+1)h = (c+1)h = 0$.

1.3 Route 3: Mahler structure

Conjecture 2 ($U(t)$ is not a class-(1) Mahler function)**.** *$U(t)$ is not a Mahler function of class (1) in the sense of Adamczewski–Faverjon 2020.*

Were Conjecture 2 established, the Adamczewski–Faverjon framework would yield a third independent route to transcendence of β_2, parallel to Propositions 2 and 4.

1.4 Joint picture

The three conditionals are logically independent. For Conjecture 1 to fail, all three hypotheses would have to fail simultaneously.

2. Structural exclusions for $U(t)$ at low complexity

The following finite linear-algebra exclusions on $u_0, \dots, u_{42}$ constrain the form of $U(t)$ at the verifiable horizon.

Proposition 5 (No low-order linear recurrence)**.** *The sequence $u_0, \dots, u_{38}$ satisfies no linear recurrence over $\mathbb{Q}$ of order ≤ 19. (Equivalently: for every $k \leq 19$, the system $u_n = \sum_{j=1}^{k} c_j u_{n-j}$ has empty rational solution set on the available terms.)*

Proposition 6 (No nonlinear or modular structure at low complexity)**.** *On $u_0, \dots, u_{42}$:*

(i) *No D-finite recurrence with $k \leq 9$ and $m \leq 7$ (Proposition 3).*

(ii) *No nonlinear polynomial relation $\sum_\alpha q_\alpha(n)\, u_n^{\alpha_0} \cdots u_{n-w}^{\alpha_w} = 0$ with lookback $w \leq 5$, total u-degree $|\alpha| \leq 3$, and coefficient degree ≤ 2.*

(iii) *No non-trivial modular linear recurrence at any modulus $m \in \{3, 4, \dots, 101\}$ except $m = 2$, where the order-2 Fibonacci recurrence $u_d \equiv u_{d-1} + u_{d-2} \pmod 2$ holds throughout the tested range.*

Proposition 7 (Coefficient-height gap)**.** $\log u_d = c \cdot d + o(d)$ *with* $c \approx 0.404$ *(linear regression on $d \in [25, 42]$, $R^2 \geq 0.999$). Consequently u_d is not the value sequence of any k-automatic integer sequence (Bell–Coons / Adamczewski–Bell height-gap theorem) and is not the coefficient sequence of any Mahler function of bounded complexity.*

Proposition 8 (Mod-2 algebraicity)**.** $u_d \bmod 2$ *is purely 3-periodic: $u_d \equiv 1$ for $d \not\equiv 0 \pmod 3$ or $d \leq 2$, and $u_d \equiv 0$ otherwise. Equivalently $u_d \equiv u_{d-1} + u_{d-2} \pmod 2$ for $d \geq 2$. Hence $U(t) \in \mathbb{F}_2[[t]]$ is algebraic over $\mathbb{F}_2(t)$ via a 4-state finite automaton.*

Caveat. *This is a characteristic-2 statement and does not lift to algebraicity over $\mathbb{Q}(t)$.*

3. Universality class: scope and known examples

The mechanism that proves universality in the main paper (the universality theorem of [7]) (a cyclic-ideal identification $I_A \cap I_B = \mathbb{Z}[Q] \cdot h$ with explicit central generator h inside the linear quotient Q of a free product $W = A * B$) admits a natural class of free-product reflection systems.

Conjecture 3 (Open question: scope of the cyclic-ideal mechanism)**.** *Characterise the pairs (W, ρ) for which the cyclic-ideal mechanism of the universality theorem of [7] applies and yields universality of the orbit-growth sequence across the family of geometric realisations.*

Two examples are currently known.

Example A: right triangles. $A = V_4$, $B = \mathbb{Z}/2$, $Q = D_\infty \times \mathbb{Z}/2$, $h = (c-1)(Y-1)$. This is the main paper's the universality theorem of [7].

Example B: hyperbolic two-right-angle quadrilaterals. $A = B = V_4$, $W = V_4 * V_4$, $Q = D_\infty \times \mathbb{Z}/2$, $h = c - 1$ (no $(Y-1)$ factor needed: centrality of c alone suffices when both factors of the free product contain c).

Cases where the mechanism is structurally blocked. Acute and obtuse triangle reflection groups have no central involution playing the role of c. Equilateral and other triangle groups with all finite Coxeter labels lack the free-product structure. The rectangle group $D_\infty \times D_\infty$ is a direct product, not a free product. Right-corner orthoschemes in dimension $n \geq 3$ carry cross-relations $(R_i R_j)^2 = 1$ between generators of distinct free-product factors, again destroying the $W = A * B$ structure; universality in this cyclic-ideal sense is therefore specific to dimension 2.

4. Structural reading of universality (algebraic-channel BFS)

Fix $\rho_{a,b} : W \to \mathrm{Aff}(\mathbb{R}^2)$. Let Y denote the linear part of $\rho_{a,b}(R_2)$ and v_2 its translation part. The identity $Yv_2 = -v_2$ holds for every (a, b) and is the image of a single canonical word $w_\star \in W$. Let N_{alg} be the normal closure of $w_\star$ in W.
Numerical observation 1 (Empirical algebraic form of universality). Through depth 22, on the four test right triangles $(3, 4), (5, 12), (1, 2), (7, 24)$, the equivalence relation on canonical Coxeter words generated by $Yv_2 + v_2$ in $\mathbb{Z}[D_\infty]$ exactly recovers the affine equivalence relation. Equivalently: $N(T) = N_{\mathrm{alg}}$ on the depth-≤ 22 filtration for every tested T.

A five-channel parallel BFS verifies this with cumulative state count 118 938 at depth 22 in every channel. Reading: the eight length-10 affine relations of the symbolic-verification theorem of [7] are not sporadic length-10 coincidences but the first eight consequences of the single quadratic relation $Yv_2 + v_2 = 0$, made visible by the canonical-word filtration.

5. Empirical evidence against finite confluent rewriting

Knuth–Bendix experiment. GAP `KBMAG` ran Knuth–Bendix completion on W together with the eight length-10 affine relations of the symbolic-verification theorem of [7]; the procedure produced 32 730 derived rules in approximately 8 seconds of `kbprog` computation before hitting the equation-count cap, never reaching confluence. This is empirical evidence (not a proof) that W/N has no finite confluent rewriting system in shortlex on the standard generators.

Antolín–Ciobanu route (structurally blocked). The Antolín–Ciobanu rationality theorem for conjugacy-growth series of non-elementary acylindrically hyperbolic groups does not apply to $U(t)$: the quotient $W/\ker(\rho_{\mathrm{abs}})$ has image inside $Q \ltimes M \subseteq (\mathbb{Z} \wr \mathbb{Z}) \rtimes V_4$, a finitely-presented virtually metabelian group, which is not acylindrically hyperbolic.

6. Class A and Class B in 3D

The 3D right-corner orthoscheme reflection groups fall into three faithfulness regimes by leg-symmetry, the unconditional Class C case being treated in the main paper (Theorem the Class C faithfulness theorem of the n-dimensional companion paper).

Class A ($a = b = c$, S_3-symmetric, polynomial growth). The $(1, 1, 1)$ cube-corner orthoscheme is the fundamental chamber of the affine Weyl group $\widetilde{C}_3$. For $n \geq 1$, $u_n = (8n^2 + c_{n \bmod 5})/5$ with $c = (10, 12, 13, 13, 12)$, with asymptotic $u_n \sim (8/5)n^2$. Order-7 linear recurrence with characteristic polynomial $(x - 1)^3\Phi_5(x)$, obtained from the parabolic Poincaré series $W_A(t) = N(t)/((1 - t)^3\Phi_5(t))$ with $N(t) = 2 + 3t^2 + 3t^3 + 3t^4 + t^5 + 4t^6$. This is a clean restatement of standard affine-Weyl-group material (Bourbaki Ch. VI §4; Humphreys §4.7) and is recorded here for completeness.

Class B (exactly two equal legs, intermediate exponential rate). Through depth 18 the sequence on $(1, 1, 2)$ is $1, 4, 9, 18, 35, 67, 129, 249, 480, 925, 1783, 3437, 6625, 12770, 24591, 47298, 90948, 174939, 33$ with empirical ratio settling around ≈ 1.92. No order-≤ 9 linear recurrence over $\mathbb{Q}$ fits the depth-18 data. No closed form is known.

7. Other deferred material

The following items were deferred from the main paper to this supplement and remain available in earlier drafts of the project:

- Numerical and structural analysis of the representation gap β_n vs. r_n across dimensions.
- Collision-class enumeration to depth 30, the structural sequence battery, the $\sqrt{3}$-asymptotic of $\delta_d - 2z_d$, and the crossover $\delta_d > u_d$ for $d \geq 24$.
- $\{3, 4, 5\}$-cascade observation for maximal outerplanar graphs on $n \leq 15$ vertices, the slack/denominator/preci analysis, and the cascade-existence conjecture.
- The standard 3-simplex remark and merged length-10-relation cascades from Pythagorean triples sharing a common edge length.
- Per-script hardware envelopes and runtime tables for the BFS implementations.

These items are unchanged from earlier supplementary drafts and are recorded here as a placeholder; consult the project repository for the detailed records.

References

References to literature are as in the main paper. The additional references invoked by the conditional routes above are:

- G. Christol, *Ensembles presque périodiques k-reconnaissables*, TCS 9 (1979), 141–145.
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[7] V. Bonfioli, *Transcendence of the right-triangle reflection growth series*, companion paper; code and formalization at `github.com/ElVec1o/pythagorean-reflection-sequence`.